

**MATHEMATICAL MODELLING OF REAL-WORLD
DYNAMICAL SYSTEMS USING ORDINARY
DIFFERENTIAL EQUATIONS**

Prasha Gujrati

ABSTRACT

Mathematical modelling provides a systematic framework for representing real-world phenomena using mathematical equations. This project develops and analyses a collection of dynamical models based on ordinary differential equations (ODEs), following the calculations contained in the accompanying Wolfram Mathematica notebook. The applications include population growth and decay, technology adoption, density dependence, harvesting, lake pollution, seasonal environmental variation, drug assimilation, epidemic transmission, battle dynamics, and predator–prey interactions. Analytical solutions are used where closed forms are available, while numerical integration is used for nonlinear and coupled systems. Twenty-two distinct computational plots from the notebook are reproduced and interpreted. Together, the models demonstrate growth, decay, saturation, equilibrium, oscillation, transfer between compartments, epidemic peaks, mutual attrition, ecological cycles, and parameter sensitivity.

Keywords: mathematical modelling; ordinary differential equations; dynamical systems; Mathematica; numerical methods; population dynamics; epidemiology; predator–prey systems.

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1 Introduction

Many real-world quantities change continuously with time. Population size changes because of births and deaths; pollutants enter and leave environmental systems; medicines are absorbed and eliminated; diseases spread between individuals; opposing forces interact; and predator and prey populations influence one another. Such processes can be represented by studying their rates of change.

For a state variable $y(t)$, a general first-order ODE is

$$\frac{dy}{dt} = f(t, y). \quad (1)$$

The function f describes the mechanisms that increase or decrease the state. A model is completed by specifying parameters and an initial condition such as $y(0) = y_0$.

The modelling process used throughout this report is

real-world assumptions $\longrightarrow$ mathematical model $\longrightarrow$ ODE $\longrightarrow$ solution $\longrightarrow$ graph $\longrightarrow$ interpretation.

The objective is not merely to calculate a trajectory, but to connect the shape of that trajectory to the mechanisms represented in the differential equation.

2 Objectives

The major objectives are:

- to formulate ODEs from real-world assumptions and balance laws;
- to solve single equations and coupled systems analytically or numerically;
- to implement and inspect the models computationally;
- to visualise every distinct case contained in the Mathematica notebook;
- to interpret growth, decay, equilibrium, saturation, oscillation and interaction;
- to examine the influence of parameters and initial conditions; and
- to recognise modelling limitations and numerical diagnostics.

3 Methodology and Computational Environment

For each application, the state variables were defined, simplifying assumptions were stated, a rate equation was formulated, and initial conditions were supplied. Mathematica's `DSolve` and `NDSolve` were used in the source notebook. The notebook solutions and equations were then evaluated on dense time grids to produce publication-ready versions of all 22 distinct graphs.

The principal Wolfram Language functions represented in the notebook are `DSolve`, `NDSolve`, `Plot`, `Table`, `Evaluate`, `PlotRange`, and `PlotLegends`. Symbolic methods are most useful for linear or

separable equations. Numerical methods are essential for most nonlinear, epidemic, combat, and ecological systems.

Notebook consistency note

The source notebook contains several typographical or implementation inconsistencies. In the exponential case, the displayed result corresponds to $P(0) = 150$ even though one input line uses a derivative condition. In the harvesting case, $h = 0.9$ is declared but omitted from the entered ODE. In the second seasonal-flow case, multiplication is used where the surrounding balance-law discussion indicates inflow minus outflow. For these cases, the figures use the stated initial value or intended balance-law model. The repeated-pill graphs, however, retain the equations exactly as entered in the notebook so that their unusual behaviour remains visible.

4 Population Models

4.1 Case 1: Exponential Population Growth

Let $P(t)$ denote population in millions. With a constant proportional growth rate $r = 0.02$,

$$\frac{dP}{dt} = 0.02P, \quad P(0) = 150. \quad (2)$$

The analytical solution is

$$P(t) = 150e^{0.02t}. \quad (3)$$

Consequently,

$$P(10) = 150e^{0.2} \approx 183.21 \text{ million.}$$

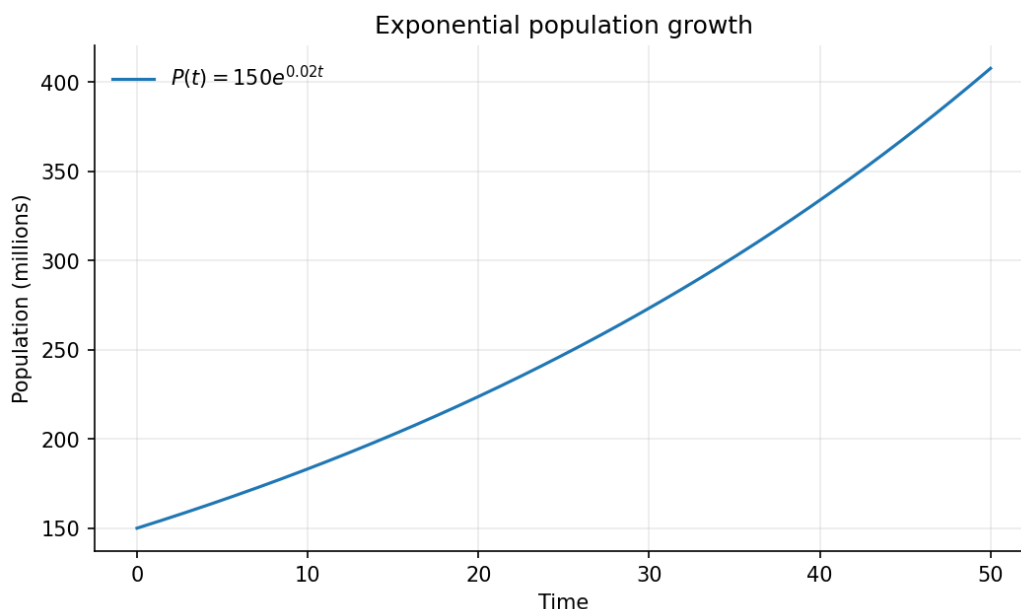


Figure 1: Exponential population growth over the notebook's plotting interval.

The curve is increasing and convex. A constant percentage growth rate therefore produces an increasing absolute increment. This model is useful over short periods but cannot represent long-run resource limitations.

4.2 Case 2: Technology Adoption

Let $N(t)$ be the number of teachers who have adopted a technology. The notebook uses

$$\frac{dN}{dt} = 0.5N \left(1 - \frac{N}{10000} \right), \quad N(0) = 100. \quad (4)$$

Its closed-form solution is

$$N(t) = \frac{10000}{1 + 99e^{-0.5t}}. \quad (5)$$

The time at which adoption reaches 80% is obtained from $N(t) = 8000$:

$$t = \frac{1}{0.5} \log(396) \approx 11.96.$$

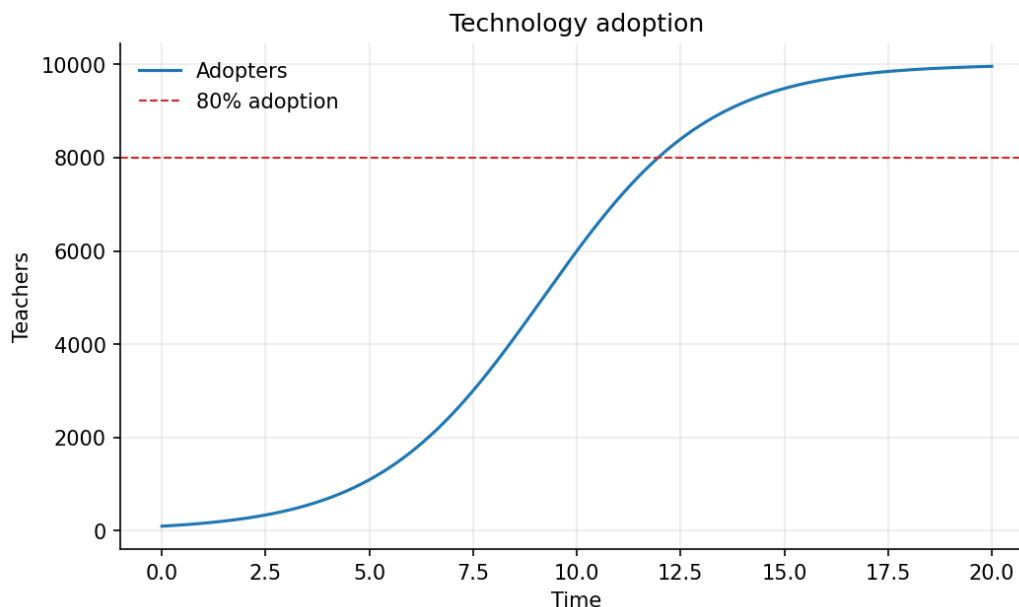


Figure 2: Logistic technology adoption and the 80% adoption threshold.

The S-shaped curve contains an initial slow phase, a rapid diffusion phase, and a saturation phase. Unlike exponential growth, the logistic equation recognises that the potential adopter population is finite.

4.3 Case 3: Density-Dependent Population Growth

The notebook models a population using

$$\frac{dy}{dt} = 0.2y - 0.001y^2, \quad y(0) = 100. \quad (6)$$

The equilibria satisfy $y(0.2 - 0.001y) = 0$, giving $y = 0$ and $y = 200$. The positive equilibrium is stable.

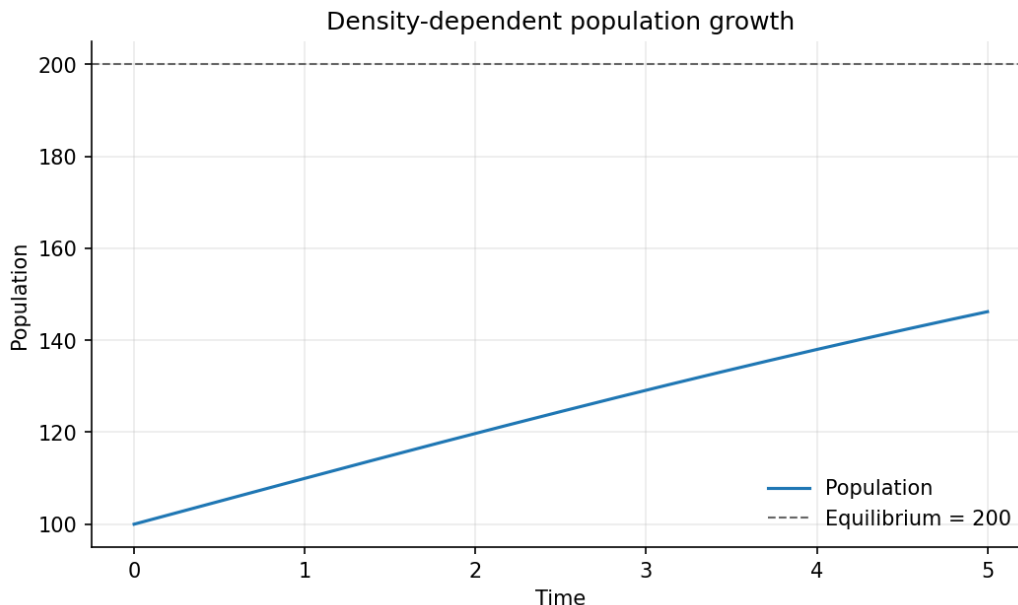


Figure 3: Density-dependent growth toward the stable population of 200.

The quadratic term represents competition for finite resources. Growth is positive below 200 and becomes progressively slower as the population approaches its equilibrium.

4.4 Case 4: Population Growth with Harvesting

The intended harvested logistic model is

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right) - H, \quad (7)$$

with $r = 1$, $K = 10$, $H = 0.9$, and $N(0) = 2$. The equilibria solve

$$N \left(1 - \frac{N}{10} \right) - 0.9 = 0,$$

which gives $N = 1$ and $N = 9$. The upper equilibrium is stable, whereas the lower equilibrium is an unstable harvesting threshold.

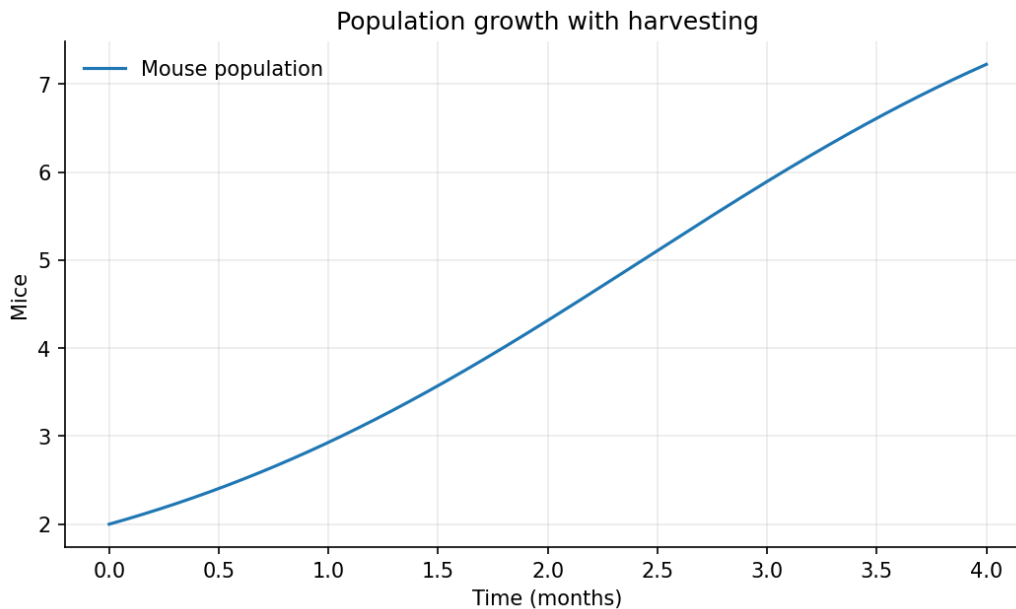


Figure 4: Mouse population under logistic growth and constant harvesting.

The graph illustrates a central resource-management principle: a fixed harvest is sustainable only if the population begins above the critical threshold and the removal rate does not exceed the system's regenerative capacity.

5 Lake Pollution Models

5.1 Case 5: Constant Inflow Concentration

Let $C(t)$ be pollutant concentration, C_{in} the incoming concentration, F the flow rate, and V the lake volume. Conservation of mass gives

$$V \frac{dC}{dt} = FC_{\text{in}} - FC, \quad (8)$$

or

$$\frac{dC}{dt} = \frac{F}{V}(C_{\text{in}} - C). \quad (9)$$

The notebook uses $F = V = 48$, $C_{\text{in}} = 3$, and initial values $C(0) = 1, 2, 3, 4, 5$. Hence

$$C(t) = 3 + (C_0 - 3)e^{-t}. \quad (10)$$

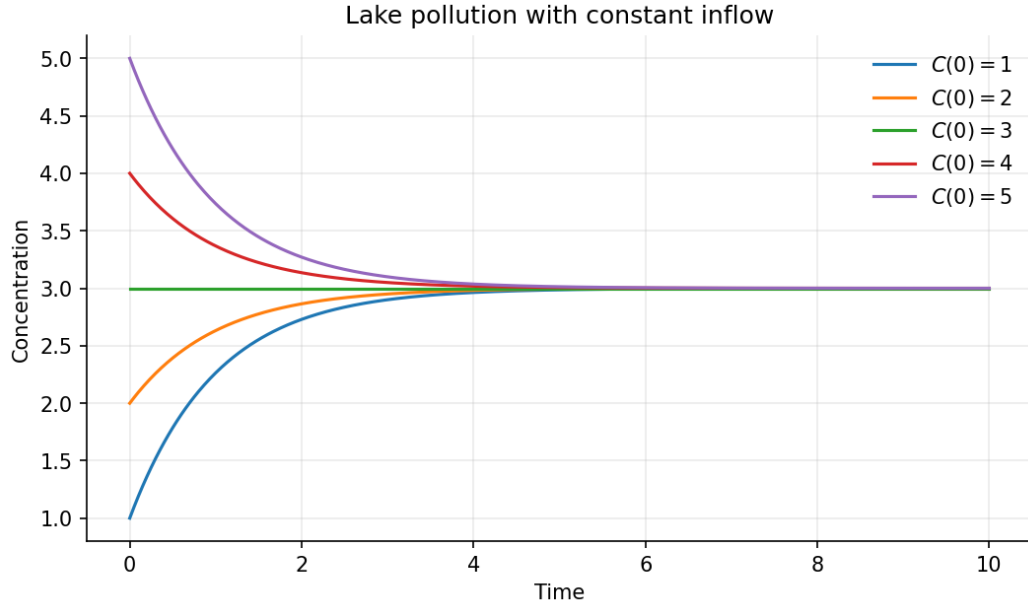


Figure 5: Lake concentration for five initial conditions and constant incoming concentration.

Every trajectory converges to the inflow concentration. Initial concentrations below 3 rise, those above 3 fall, and $C(0) = 3$ is already at equilibrium.

5.2 Case 6: Seasonally Varying Removal

The first seasonal cell in the notebook uses

$$\frac{dC}{dt} + \frac{F}{V} [1 + \varepsilon \cos(2\pi t)] C = 0, \quad (11)$$

with $F/V = 3$ and $\varepsilon = 0.01$. The periodic factor modulates the decay rate.

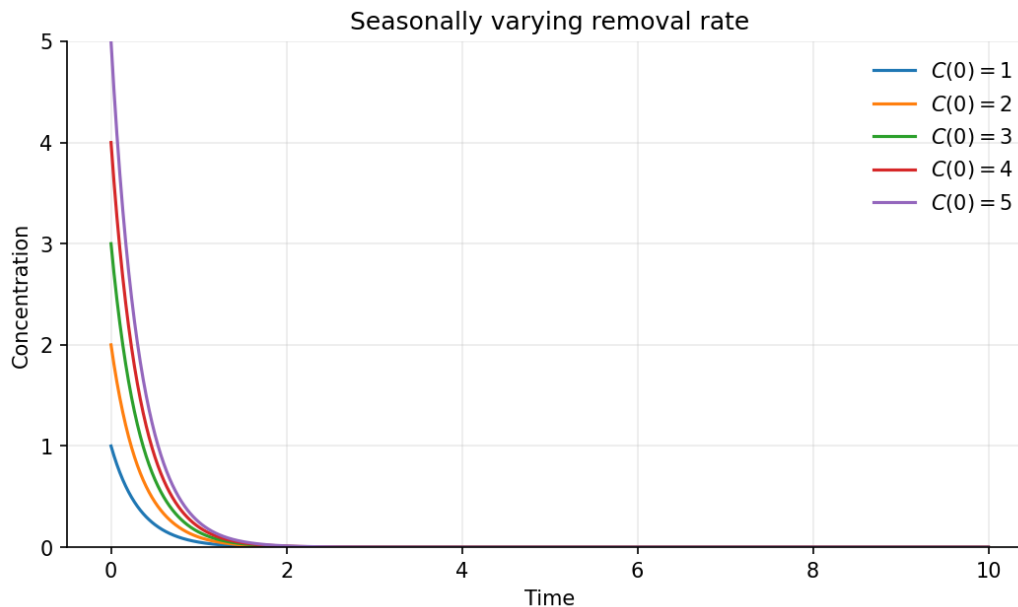


Figure 6: Seasonally modulated decay for five initial pollutant concentrations.

All trajectories decay rapidly. The seasonal effect is small relative to the mean removal rate, so it appears as a slight modulation rather than a large sustained oscillation.

5.3 Case 7: Periodic Pollutant Input

For the second seasonal case, the stated incoming concentration is

$$C_{\text{in}}(t) = 10 + 10 \cos(2\pi t). \quad (12)$$

Using the balance law with $F/V = 1$ gives

$$\frac{dC}{dt} = C_{\text{in}}(t) - C. \quad (13)$$

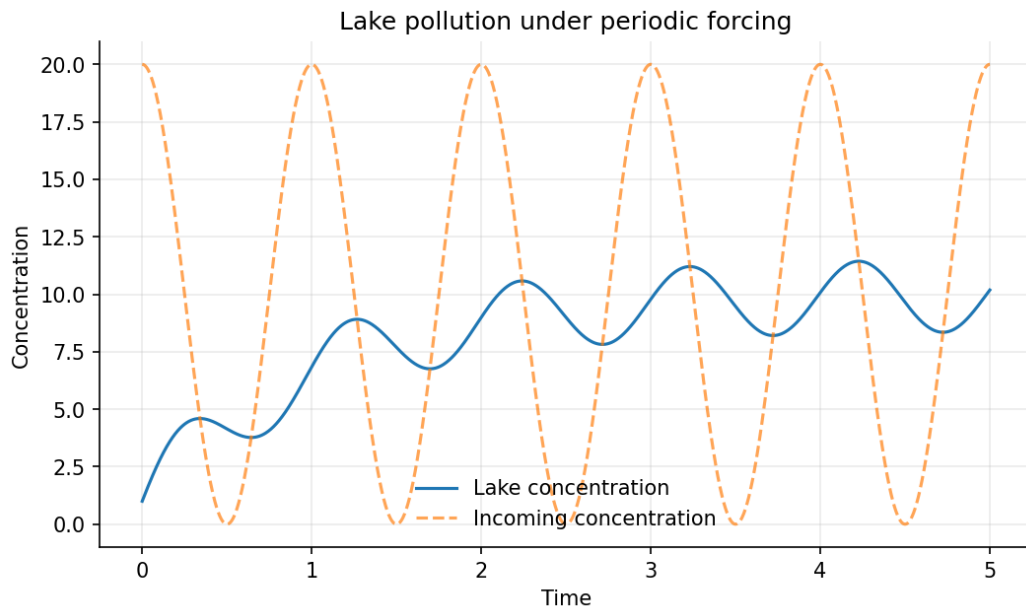


Figure 7: Lake response to periodic incoming pollutant concentration.

The lake concentration follows the forcing with reduced amplitude and a time delay. This demonstrates the filtering effect of environmental storage: a well-mixed lake smooths rapid variation in its input.

6 Drug Assimilation Models

6.1 Compartmental Structure

Let $x(t)$ be the amount of drug in the gastrointestinal (GI) tract and $y(t)$ the amount in the bloodstream. For a single pill,

$$\frac{dx}{dt} = -k_1x, \quad \frac{dy}{dt} = k_1x - k_2y, \quad (14)$$

with $x(0) = 20$ and $y(0) = 0$. The GI amount decays monotonically. The blood amount first rises while absorption dominates and later falls when elimination dominates.

6.2 Case 8: Single Cold Pill, Slow Elimination

The first parameter set is $k_1 = 1.30$ and $k_2 = 0.13$.

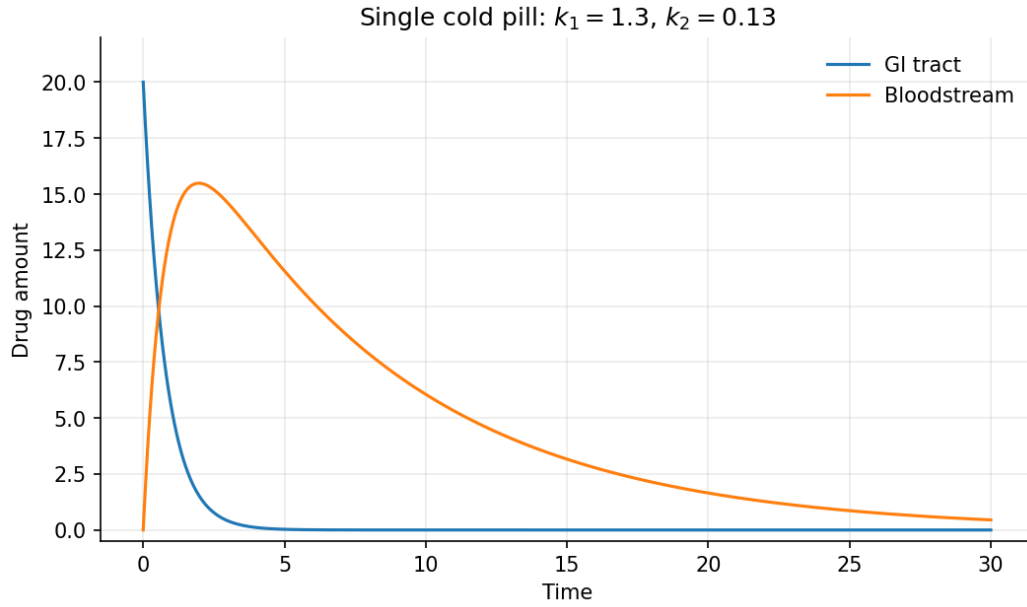


Figure 8: Drug transfer for a single pill with $k_1 = 1.30$ and $k_2 = 0.13$.

The bloodstream curve rises to a distinct peak and then declines slowly. The tenfold difference between absorption and elimination rates creates rapid entry into the blood followed by a longer clearance phase.

6.3 Case 9: Single Cold Pill, Rapid Elimination

The notebook also evaluates $k_1 = 1$ and $k_2 = 3$.

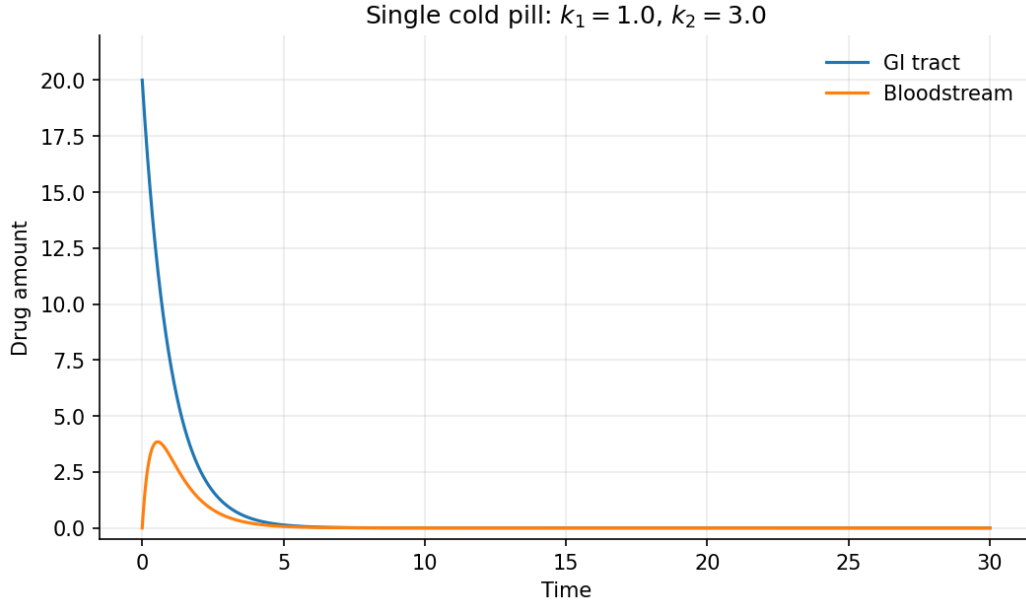


Figure 9: Drug transfer for a single pill with $k_1 = 1$ and $k_2 = 3$.

Because elimination is faster than absorption, the blood peak is lower and occurs early. This comparison shows how pharmacokinetic parameters control both peak exposure and residence time.

6.4 Repeated or Continuous Drug Input

The repeated-input notebook cells use

$$\frac{dx}{dt} = m - k_1x, \quad \frac{dy}{dt} = k_1x - k_2y = (k_1 - k_2)x, \quad (15)$$

with $m = 1$ and $x(0) = y(0) = 0$. Notice that the second loss term contains x , not y . This differs from the usual compartmental clearance equation and explains the unbounded blood trajectories in Cases 10 and 11.

6.5 Case 10: Course of Pills, $k_1 = 0.69$, $k_2 = 0.023$

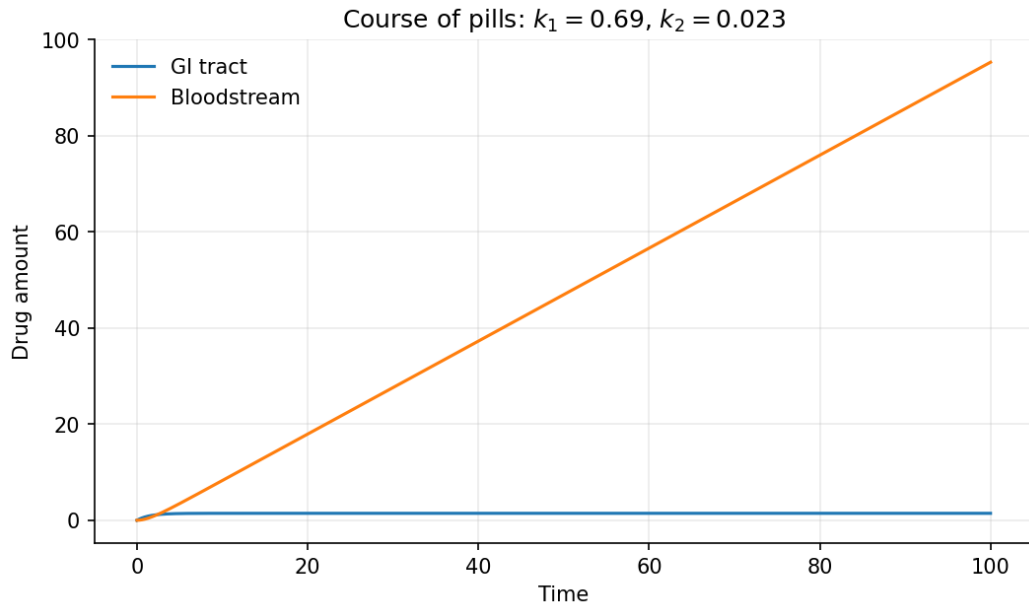


Figure 10: Repeated-input notebook model with $k_1 = 0.69$ and $k_2 = 0.023$.

The GI compartment approaches the steady amount m/k_1 , while the blood variable grows approximately linearly after the initial transient because $(k_1 - k_2)x$ remains positive.

6.6 Case 11: Course of Pills, $k_1 = 5$, $k_2 = 1$

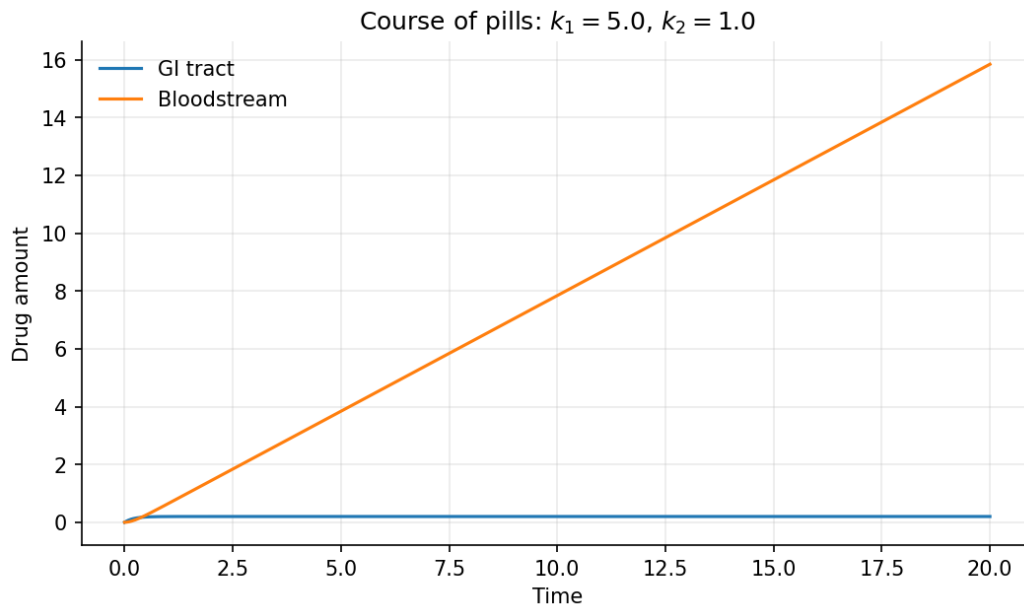


Figure 11: Repeated-input notebook model with $k_1 = 5$ and $k_2 = 1$.

Faster GI equilibration produces an almost immediate plateau in $x(t)$. The positive difference $k_1 - k_2$ again produces continued growth in $y(t)$.

6.7 Case 12: Course of Pills, $k_1 = k_2 = 1$

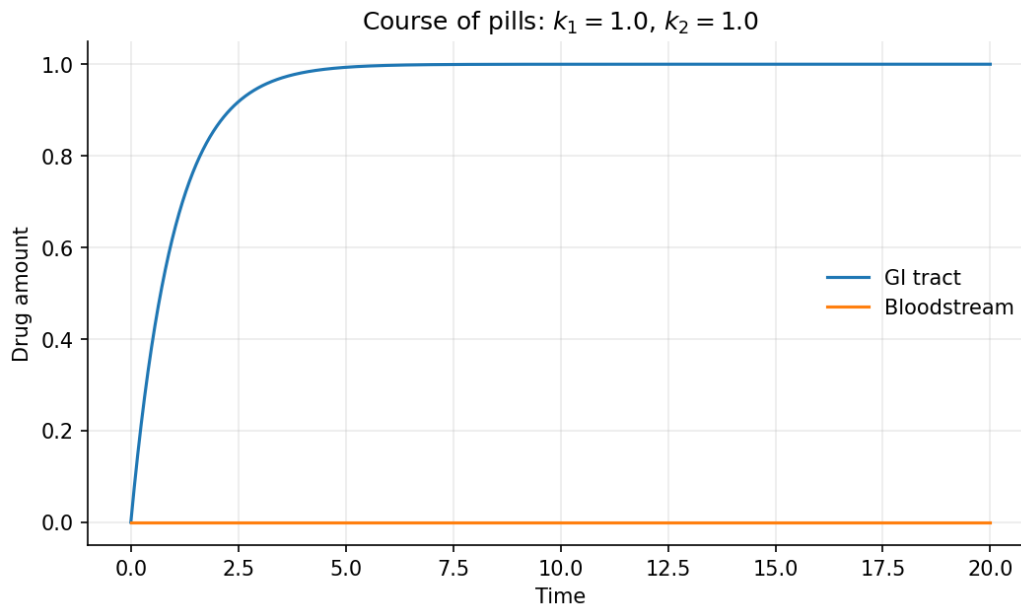


Figure 12: Repeated-input notebook model with equal transfer coefficients.

Here $dy/dt = 0$, so the blood variable remains at its initial value. This limiting case makes the consequence of the notebook's entered equation especially clear. In a standard repeated-dose model, a term $-k_2y$ would instead create a nonzero steady-state blood level.

7 Epidemic Models

7.1 Case 13: Basic Influenza SIR Model

The susceptible–infected–recovered model is

$$\frac{dS}{dt} = -\beta SI, \quad (16)$$

$$\frac{dI}{dt} = \beta SI - \gamma I, \quad (17)$$

$$\frac{dR}{dt} = \gamma I. \quad (18)$$

The notebook uses $\beta = 2.18 \times 10^{-3}$, $\gamma = 0.44$, and $(S(0), I(0), R(0)) = (762, 1, 0)$.

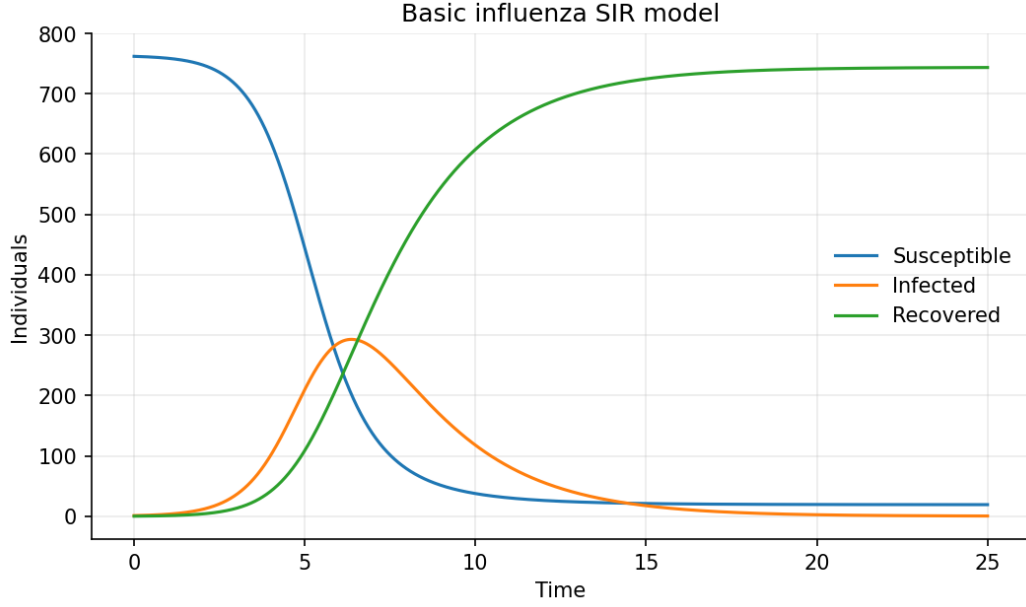


Figure 13: Susceptible, infected and recovered populations in the basic influenza model.

The infected population grows when $\beta S > \gamma$, reaches a peak when $S = \gamma/\beta$, and then declines. The recovered population is monotone increasing and the total population remains constant.

7.2 Case 14: Disease Contagious for Life

If there is no recovery, the model reduces to

$$\frac{dS}{dt} = -\beta SI, \quad (19)$$

$$\frac{dI}{dt} = \beta SI, \quad (20)$$

with $\beta = 0.002$, $S(0) = 500$, and $I(0) = 1$.

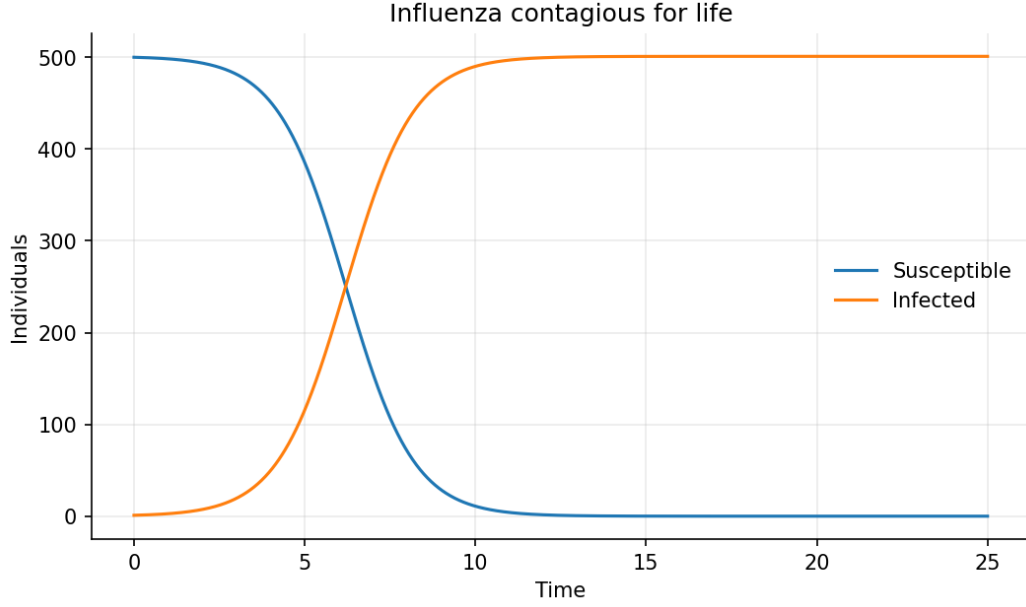


Figure 14: SI epidemic in which infected individuals remain contagious.

The susceptible and infected populations exchange mass while $S + I$ remains constant. With no recovery mechanism, infection persists and eventually occupies almost the entire closed population.

7.3 Case 15: Influenza with a Carrier Compartment

Let $C(t)$ represent carriers. The notebook model is

$$S' = -\beta SI - \varepsilon \beta SC, \quad (21)$$

$$I' = \beta SI + \varepsilon \beta SC - \gamma I, \quad (22)$$

$$C' = q\gamma I, \quad (23)$$

$$R' = (1 - q)\gamma I, \quad (24)$$

where $\beta = 2.18 \times 10^{-3}$, $\gamma = 0.44$, $q = 0.4$, $\varepsilon = 0.5$, and $(S, I, C, R)(0) = (762, 0, 1, 0)$.

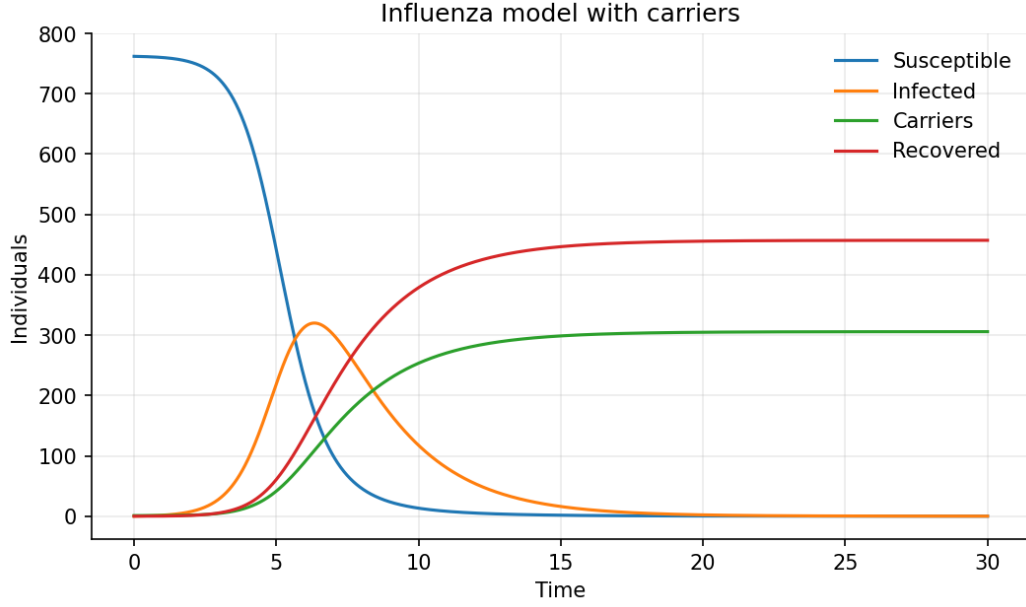


Figure 15: Influenza dynamics with susceptible, infected, carrier and recovered compartments.

The carrier compartment provides a continuing transmission pathway. This extension demonstrates how adding a biologically meaningful transition can substantially alter epidemic persistence and final compartment sizes.

8 Battle Models

8.1 Case 16: Basic Battle Model

The notebook's basic Lanchester-type system is

$$\frac{dR}{dt} = -a_1 B, \quad (25)$$

$$\frac{dB}{dt} = -a_2 R, \quad (26)$$

with $a_1 = 0.0544$, $a_2 = 0.0106$, $R(0) = 66454$, and $B(0) = 18274$.

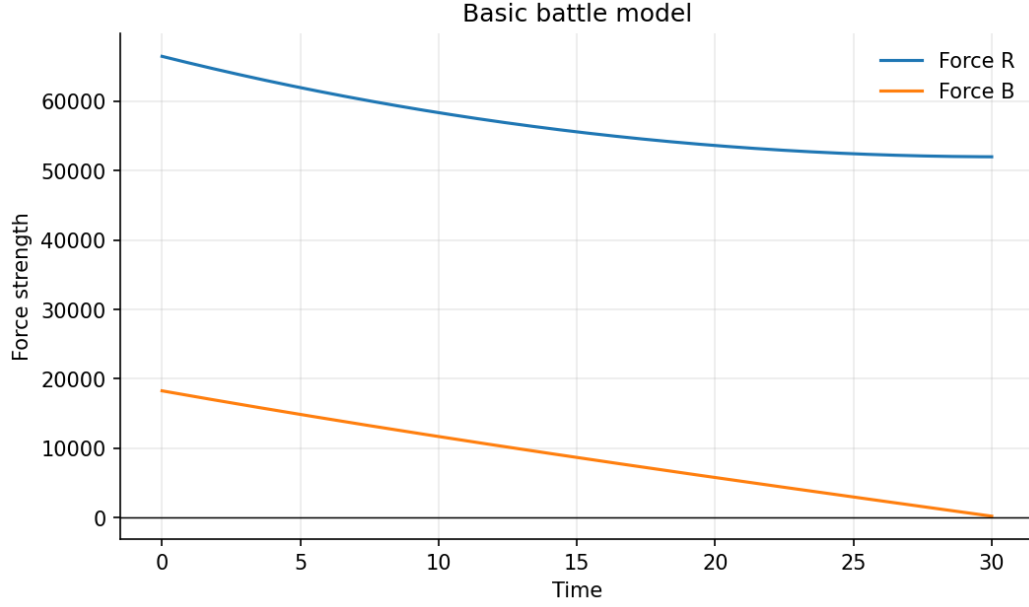


Figure 16: Basic mutual-attrition battle model.

Both forces decline because losses are caused by the opposing force. Once a mathematical trajectory crosses zero, the model has left its physically meaningful region; the crossing time should therefore be interpreted as the end of the engagement rather than as a negative force strength.

8.2 Case 17: Jungle Warfare

The jungle-warfare equations are

$$R' = -4B, \quad (27)$$

$$B' = -4 \times 10^{-4}RB, \quad (28)$$

with $R(0) = 500$ and $B(0) = 150$.

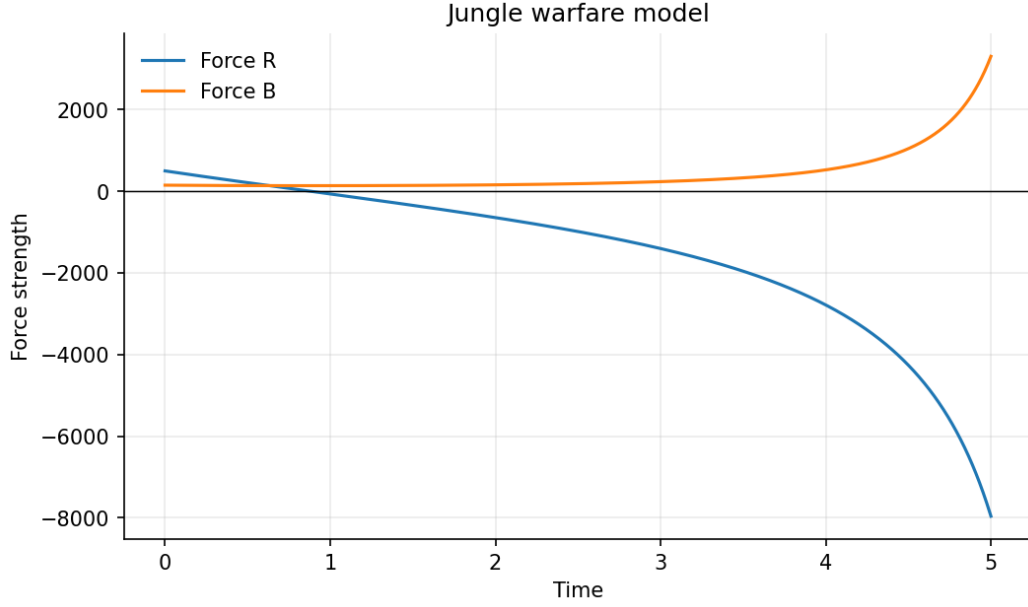


Figure 17: Jungle-warfare model with asymmetric attrition mechanisms.

One side experiences losses proportional to the opposing strength, while the other experiences interaction-dependent losses. The graph highlights the strong sensitivity of battle outcomes to both equation structure and parameter scale.

8.3 Case 18: Long-Range Weapons

The long-range model uses

$$R' = -0.05BR, \quad (29)$$

$$B' = -0.004RB, \quad (30)$$

with $R(0) = 100$ and $B(0) = 50$.

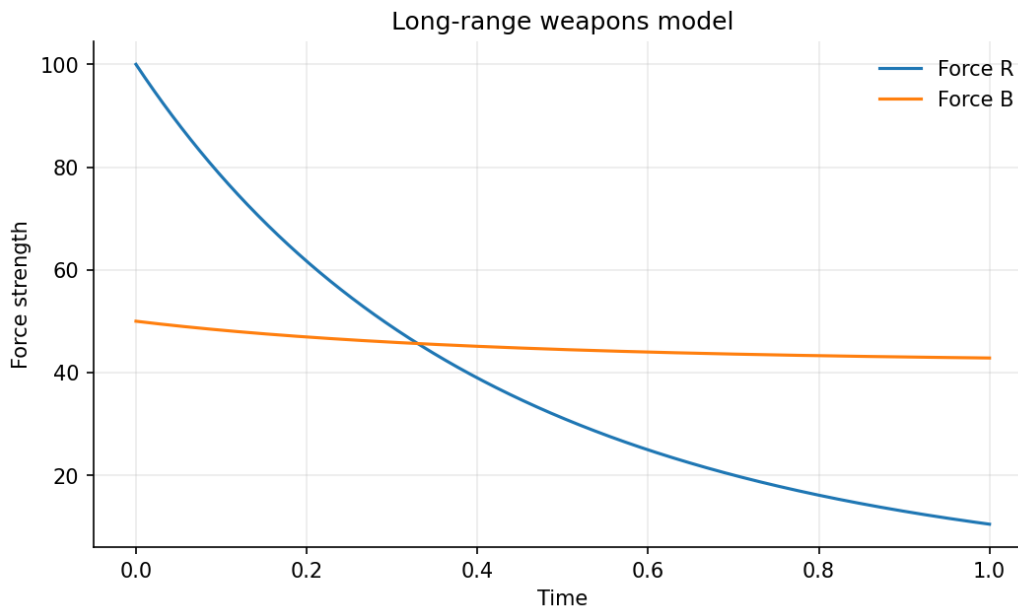


Figure 18: Long-range weapons model over the plotted interval $0 \leq t \leq 1$.

Both rates contain the product RB , producing rapid nonlinear change early in the engagement. The source notebook reports a numerical step-size warning when it attempts a much longer interval. This diagnostic indicates that the solver is being pushed into a region with extreme scales or poor physical relevance. The plotted interval is the shorter interval explicitly requested by the notebook's `Plot` command.

9 Predator–Prey Models

9.1 Case 19: Basic Lotka–Volterra Model

Let $x(t)$ be prey and $y(t)$ predators. The equations are

$$x' = x - 0.01xy, \tag{31}$$

$$y' = 0.005xy - 0.5y, \tag{32}$$

with $(x(0), y(0)) = (200, 80)$.

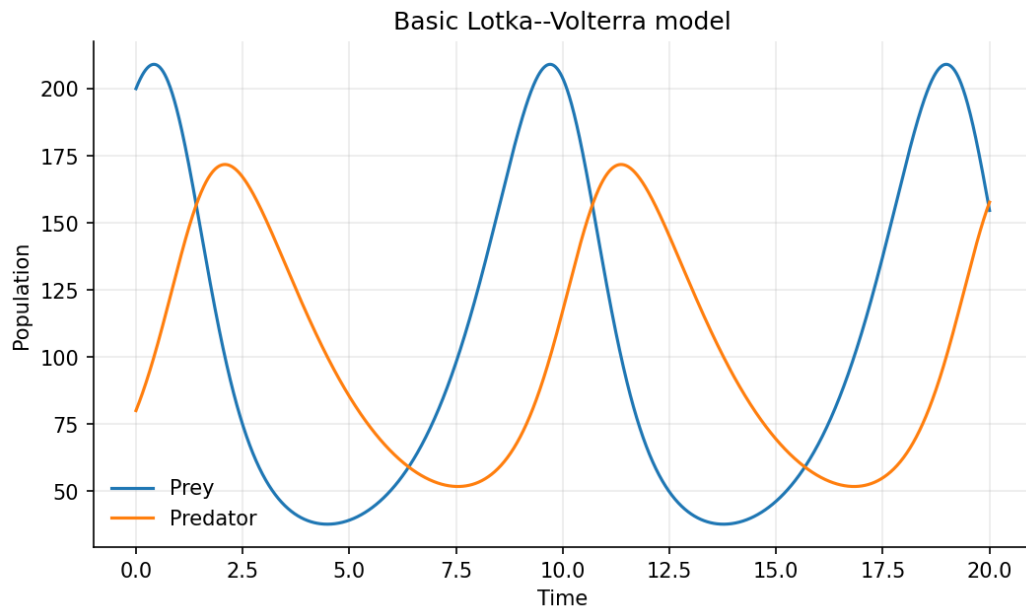


Figure 19: Oscillatory prey and predator populations in the Lotka–Volterra model.

Prey growth supports an increase in predators; increased predation then reduces prey; predator numbers subsequently fall; and prey recover. The predator peaks lag behind the prey peaks because predator growth depends on prey availability.

9.2 Case 20: Predator–Prey Model with DDT

The DDT model adds mortality rates $p_1 = p_2 = 0.1$:

$$x' = x - 0.01xy - 0.1x, \quad (33)$$

$$y' = 0.005xy - 0.5y - 0.1y. \quad (34)$$

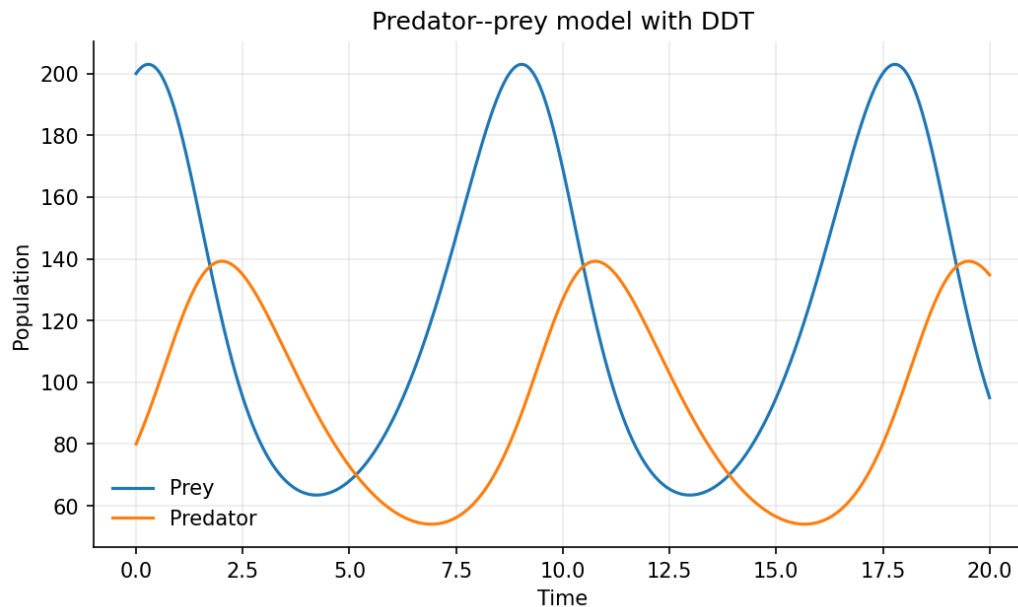


Figure 20: Predator-prey dynamics with pesticide mortality.

The intervention affects both populations and shifts the ecological balance. The resulting cycles differ in timing and amplitude from the untreated model, illustrating how an intervention directed at one part of an ecosystem can create indirect effects throughout the system.

9.3 Case 21: Density-Dependent Predator-Prey Model

The prey growth term is restricted by a carrying capacity $K = 1000$:

$$x' = x \left(1 - \frac{x}{1000} \right) - 0.01xy, \quad (35)$$

$$y' = 0.005xy - 0.5y. \quad (36)$$

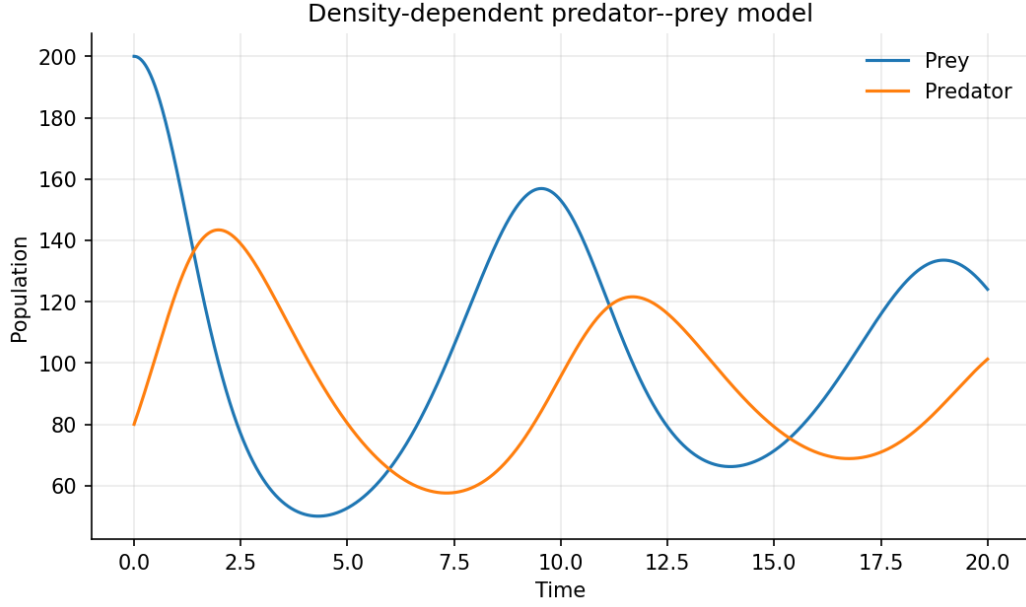


Figure 21: Predator–prey dynamics with density-dependent prey growth.

The carrying capacity prevents unlimited prey growth and changes the amplitude and damping properties of the oscillations. This is more realistic than assuming that prey grow exponentially whenever predators are absent.

9.4 Case 22: Two Prey Species and One Predator

The final notebook model contains three state variables:

$$x' = x - 0.01xz, \quad (37)$$

$$y' = y - 0.005yz, \quad (38)$$

$$z' = 0.01xz + 0.001yz - 0.5z, \quad (39)$$

with $(x(0), y(0), z(0)) = (200, 180, 100)$.

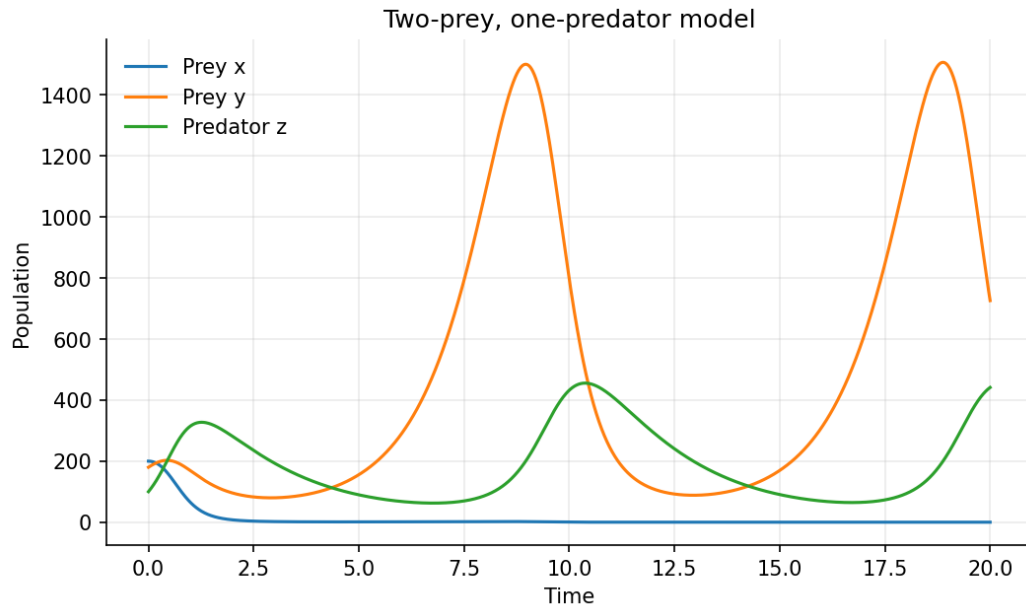


Figure 22: Dynamics of two prey populations sharing one predator.

The predator draws resources from both prey populations, while the prey species are linked indirectly through their shared predator. Increasing the system dimension creates more complicated transients and makes graphical inspection particularly valuable.

10 Comparative Analysis

Model group	Main mechanism	Typical graphical behaviour
Exponential population	Proportional growth	Increasing convex curve
Logistic and density-dependent growth	Finite capacity or competition	Saturation at equilibrium
Harvesting	Growth plus constant removal	Shifted equilibria and threshold
Lake pollution	Inflow–outflow balance	Convergence to input concentration
Seasonal pollution	Periodic coefficients or forcing	Smoothed and lagged oscillation
Drug assimilation	Transfer between compartments	Absorption peak and elimination
Epidemics	Transmission and recovery	Infection peak or persistent infection
Battle models	Mutual attrition	Depletion and possible boundary crossing
Predator–prey systems	Nonlinear biological interaction	Cycles and delayed predator response
Multi-species ecology	Shared interaction network	Complex multi-trajectory feedback

Table 1: Comparison of the principal dynamical mechanisms.

The project progresses from one-dimensional autonomous equations to non-autonomous forcing and then to coupled nonlinear systems. Each added mechanism changes the qualitative behaviour. A carrying capacity changes unbounded growth into saturation; recovery changes persistent infection into an epidemic wave; pesticide mortality changes ecological cycles; and an additional prey population creates indirect competition through a shared predator.

11 Analytical and Numerical Solutions

Analytical solutions provide explicit formulas, direct parameter dependence, and exact equilibrium behaviour. They are especially effective for exponential growth, logistic growth, linear pollution equations, and simple compartment models. Numerical solutions approximate the trajectory at selected time points and are required for most nonlinear coupled systems.

Numerical output must be interpreted critically. Solver warnings can result from stiffness, singular behaviour, excessive extrapolation, inconsistent equations, or a trajectory entering a physically meaningless region. A numerical curve is therefore evidence about the specified mathematical model, not automatic proof that the model accurately represents reality.

12 Initial Conditions and Parameter Sensitivity

Initial conditions determine the system's starting state and may change transient or long-run behaviour. In the lake model, different initial concentrations approach the same equilibrium from different directions. In epidemic models, changing the initial number infected changes the timing of the outbreak. In predator-prey systems, changing either initial population changes the phase and amplitude of the cycles.

Parameters represent mechanisms rather than arbitrary constants. Growth rates control time scale; carrying capacities set saturation levels; harvesting rates create thresholds; absorption and elimination coefficients determine drug peaks; infection and recovery rates control epidemic growth; and predation coefficients determine ecological feedback strength. Parameter estimation and sensitivity analysis are therefore essential when a model is used for prediction or policy.

13 Major Findings

1. Constant proportional growth produces exponential rather than linear change.
2. Logistic and density-dependent terms replace unlimited growth with a stable carrying level.
3. Constant harvesting can create both a sustainable equilibrium and a critical population threshold.
4. Balance laws naturally produce first-order environmental and compartmental models.
5. Periodic forcing creates a response with altered amplitude and phase.
6. Drug transfer models reproduce absorption, peak concentration, and elimination when compartment loss terms are formulated consistently.
7. Epidemic trajectories are governed by competition between transmission and removal mechanisms.
8. Nonlinear battle and ecological systems can change rapidly and are highly parameter-sensitive.
9. Predator-prey interaction can generate cycles without any external periodic input.
10. Additional species or compartments substantially increase dynamical complexity.
11. Numerical diagnostics and equation checking are indispensable parts of computational modelling.

14 Limitations

All models simplify reality. Parameters are often treated as constant; populations are assumed well mixed; spatial movement, random variation, age structure, behavioural response, migration, strategic decisions, and measurement error are generally omitted. The drug models reduce complex physiology to two compartments. The epidemic models assume homogeneous contact. The battle models omit geography, logistics, command, and tactical adaptation. The ecological models omit seasonal resources, disease, and migration.

Some source-notebook equations also differ from the verbal descriptions. Such discrepancies reinforce an important lesson: computational output must be checked against units, conservation principles, biological meaning, and the intended mathematical assumptions.

15 Conclusion

This project demonstrates the application of ordinary differential equations to a broad range of real-world dynamical systems. Beginning with exponential growth, the analysis advances through logistic and density-dependent populations, environmental balance laws, pharmacokinetic compartments, epidemic transmission, combat attrition, and nonlinear ecological interaction.

The 22 reproduced graphs show that the mathematical structure of a model determines its qualitative behaviour. Small changes—such as adding a carrying capacity, harvesting term, periodic input, recovery mechanism, carrier population, external mortality, or additional species—can produce substantially different trajectories. Mathematica provides a unified environment for symbolic solution, numerical integration, tabulation, and plotting, but the modeller remains responsible for verifying equations, selecting meaningful parameters, interpreting diagnostics, and relating mathematical results back to the real system.

Ordinary differential equations are therefore not merely theoretical objects. They are practical tools for converting assumptions into testable dynamical consequences and for understanding how interacting mechanisms shape change over time.